Data Analysis Midterm Project

Project Report — IBM HR Employee Attrition Analysis

1. Dataset Overview

This project analyzes the **IBM HR Analytics Employee Attrition dataset**, a dataset assigned for this course to be used as the basis for the midterm data analysis project. The dataset was provided directly rather than independently sourced, so this report focuses on the preparation, analysis, and dashboard work performed on it rather than on dataset selection.

Dataset Summary

Business Domain	Human Resources (Employee Attrition & Retention)
Records (Rows)	1,470 employees
Original Variables (Columns)	35
Cleaned Dataset Variables	41 (after feature engineering)
Unit of Analysis	One row per employee

Objectives of the Analysis

- Identify which employee and job-related factors are most strongly associated with attrition.
- Quantify the overall attrition rate and compare it against a defined business target.
- Examine the effect of overtime, department, job satisfaction, and marital status on turnover.
- Communicate findings through an interactive Power BI dashboard for HR stakeholders.

Why the Dataset Fits the Project Requirements

The dataset satisfies the project's structural requirements: it is a structured tabular dataset with 1,470 records (well above the 1,000-row minimum) and 35 original variables (above the 8-variable minimum). It mixes numerical variables (e.g., Age, MonthlyIncome, YearsAtCompany) with categorical variables (e.g., Department, JobRole, OverTime, MaritalStatus), and it contains a clear, measurable business metric — **Attrition** — that supports grouping, filtering, and comparison across departments, job roles, and demographic segments, making it well suited to HR analytics and retention analysis.

2. Data Cleaning

Data preparation was performed in Microsoft Excel using Power Query. The following steps were carried out:

- **Duplicate and missing-value check:** the dataset was inspected for duplicate records and null/missing values; none were found, so no rows required removal or imputation.
- **Standardizing text and formatting:** numeric rating fields (e.g., Job Satisfaction, Environment Satisfaction, Work-Life Balance, Education) were converted into readable categorical labels to make the data easier to interpret in visuals and slicers.

- **Feature engineering:** new calculated columns were created to support deeper analysis, including *Age Group*, *Satisfaction Level*, *Tenure Group*, *Income Category*, *Education Level*, *Experience Level*, *Prior Experience*, and *Commute Distance* (grouped from Distance From Home).
- **Data optimization:** non-informative columns with a single constant value across all records (*EmployeeCount*, *Over18*, *StandardHours*) were removed, since they added no analytical value.

Remaining Limitations

The dataset represents a single snapshot in time rather than a longitudinal record, so trends over time cannot be observed directly. In addition, satisfaction-related fields (Job Satisfaction, Environment Satisfaction, Relationship Satisfaction) are self-reported and therefore inherently subjective.

3. Exploratory Data Analysis

The overall attrition rate across the workforce is **16.12%** (237 of 1,470 employees), meaning roughly 1 in 6 employees left the company. Several clear patterns emerged from the analysis:

- **Overtime is the strongest driver of attrition observed:** employees who work overtime leave at a rate of **30.5%**, nearly three times higher than the **10.4%** rate for employees who do not work overtime.
- **Department differences:** Sales has the highest attrition rate at **20.6%**, followed by Human Resources at **19.0%**, while Research & Development has the lowest rate at **13.8%**.
- **Marital status matters:** single employees leave at **25.5%**, notably higher than married employees (**12.5%**) and divorced employees (**10.1%**), suggesting single employees may have fewer factors anchoring them to a location or role.
- **Job satisfaction is inversely related to attrition:** employees reporting the lowest job satisfaction (level 1) leave at **22.8%**, compared to just **11.3%** for employees reporting the highest satisfaction (level 4).
- **Recent hires are the most likely to leave:** attrition is concentrated heavily among employees with 0 years since their last promotion or short tenure, with the rate dropping off sharply after the first few years, as shown in the Years Since Last Promotion trend line.
- **Average monthly income (~$6,503) and average tenure (~7.0 years)** provide useful baselines: departments and job roles with income and tenure noticeably below these averages tend to show higher attrition, reinforcing compensation and career progression as retention levers.

4. Dashboard Summary

The purpose of the dashboard is to give HR stakeholders a single, interactive view of workforce attrition so they can quickly identify which employee segments are most at risk of leaving and take targeted action. The dashboard was built in Power BI using an IBM-inspired visual style (primary blue #0f62fe and accent orange #ff832b) with a structured grid layout for a professional, consistent appearance.

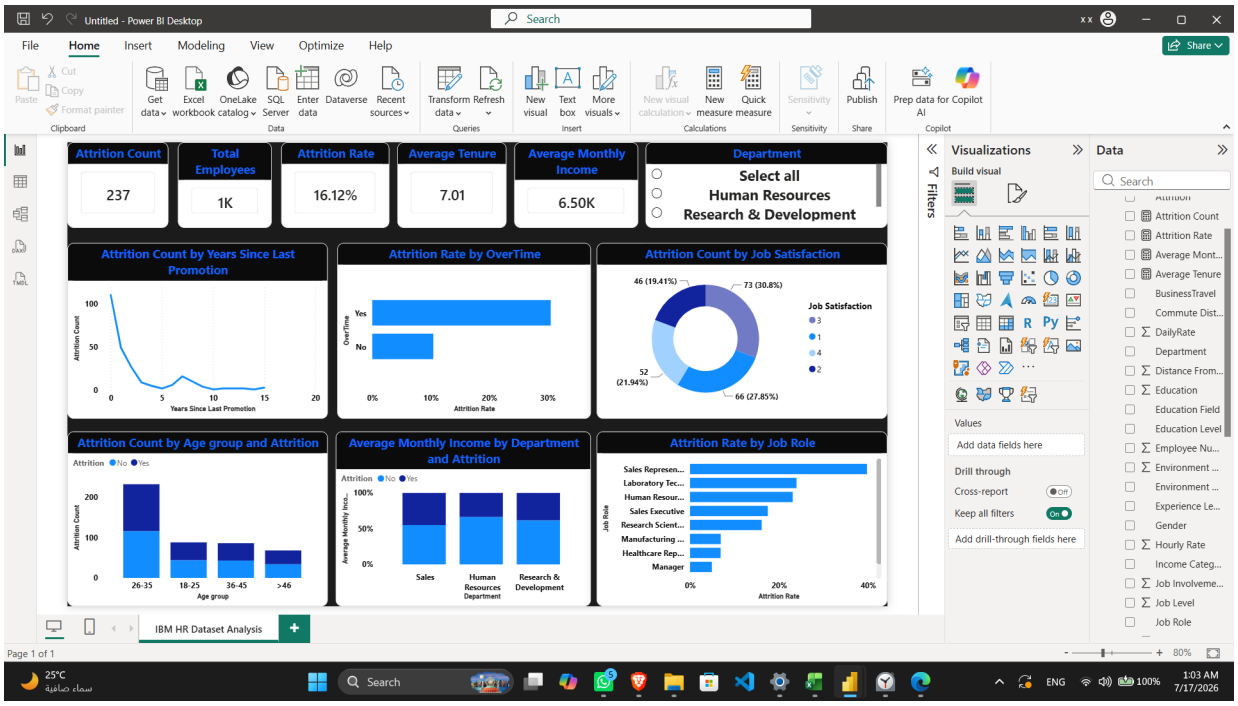


Figure 1. IBM HR Dataset Analysis dashboard, built in Power BI.

Visualizations Included

- **KPI Cards** — Attrition Count, Total Employees, Attrition Rate, Average Tenure, and Average Monthly Income, giving an at-a-glance summary of workforce health.
- **Department slicer** — allows filtering the entire dashboard by Human Resources, Research & Development, or all departments.
- **Line chart — Attrition Count by Years Since Last Promotion** — highlights that attrition is concentrated among employees who were promoted very recently or not at all.
- **Bar chart — Attrition Rate by OverTime** — directly compares turnover between employees who work overtime and those who do not.
- **Donut chart — Attrition Count by Job Satisfaction** — breaks down leavers by their reported satisfaction level.
- **Stacked column chart — Attrition Count by Age Group and Attrition** — shows workforce composition and attrition split across age bands.
- **Stacked column chart — Average Monthly Income by Department and Attrition** — compares pay levels between employees who stayed and those who left, by department.
- **Bar chart — Attrition Rate by Job Role** — ranks job roles from highest to lowest attrition rate, led by Sales Representative.

Together, these visuals answer the project's core business questions: how large is the attrition problem, which departments and roles are most affected, and which working conditions (overtime, satisfaction, promotion history) are most associated with employees leaving.

5. KPI Explanation

The dashboard's primary KPI is displayed as a Card visual: **Attrition Rate**.

What it measures	The percentage of employees who left the company out of the total workforce (Attrition Count

Target value	15%
How the target was determined	15% is a commonly cited industry benchmark for "healthy" voluntary turnover in corporate/offi
Why the target is appropriate	Turnover below roughly 10-15% is generally considered manageable and expected as part of
Result	The current Attrition Rate is 16.12%, which is above the 15% target.
Interpretation	The KPI falls slightly short of the target, indicating that attrition is marginally elevated. The gap

6. Business Insights and Recommendations

Key Insights

- **Overtime is the single biggest attrition risk factor:** employees working overtime leave at nearly three times the rate of those who do not (30.5% vs. 10.4%).
- **Sales and HR are higher-risk departments** compared to R&D, with Sales Representative standing out as the job role with the highest attrition rate.
- **Low job satisfaction and lack of recent promotion are strongly linked to leaving,** employees who have gone the longest without a promotion, and those reporting the lowest satisfaction scores, are disproportionately represented among leavers.

Recommendations

- **Monitor and cap overtime hours** in Sales and other high-overtime teams, and investigate workload distribution in roles with the highest overtime frequency.
- **Review promotion cadence,** particularly for employees who have gone several years without advancement, since delayed promotion correlates with a higher likelihood of leaving.
- **Prioritize retention initiatives for Sales Representatives and HR staff,** such as targeted compensation review or engagement surveys, given their elevated attrition rates relative to R&D.
- **Track the Attrition Rate KPI against the 15% target on a recurring basis** so HR can measure whether these interventions bring turnover back within the target range.